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Vision Revive: AI based Dehazing/De-Smoking Solution

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***Abstract*—Image dehazing is used to recuperate clear images from hazed or blurred photographs. Image dehazing should be rapid and simple to understand in order to be accessible. It may be very important for better visibility in inner extremities, such as fire risks, and be used in security and surveillance cameras as well as self-governing vehicles. A basic relic for image beautifying, it has made significant strides in recent years to recover the original image as accurately as possible without taking into account its efficacy and performance on low-end hardware. Following high-level computer vision tasks has proven difficult when it comes to single image dehazing because of the severe information deformation. The visibility of the region to be evaluated is severely reduced by atmospheric haze, endangering the accuracy of advanced-level activities. To reduce haze, different convolutional neural networks are used. Therefore, we will investigate one of the most accurate methods, which introduces a new deformable multi-head attention method for picture dehazing that is neutralized and spatially attentive for reconstruction in a restored image using fine textures. Furthermore, this enhances the deformable convolution with a spatially aware extractor with offset to concentrate on pertinent contextual information. By using edge boosting skip joins, edge features can be effectively transferred from shallow layers to deeper layers of the network.**

Keywords—Image dehazing, Real-time, Artificial Intelligence, Convolutional Neural Networks, Generative Adversarial Networks and Image processing

I. INTRODUCTION

Haze is a frequent meteorological phenomena, and photos taken in such unforgiving conditions will suffer from issues with colour and detail deterioration, which will unavoidably impair the precision of later high-level vision activities, such as autonomous driving, video surveillance, and image identification [1]. Image dehazing aims to remove haze, turn cloudy images back into clear ones, and enhance the image's visual quality so that computer vision tasks can be carried out in foggy conditions [2]. Since artificial intelligence (AI) has made significant strides in image processing methods, dehazing and de-smoking are becoming indispensable applications of AI. The high-resolution cameras used to shoot outdoor photos were also affected by weather conditions such as fog, rain, mist, smoke, haze, and snow. The visual or video quality deteriorates as a function of the current atmospheric or weather conditions. The ambient light from outside also affects the colour and contrast of the images [3]. Through the intelligent removal of smoke and haze from photos and videos, our work can produce sharper, more detailed images that can significantly improve performance on tasks like tracking, object detection, and scene interpretation. Furthermore, our paper guarantees real-time dehazing and de-smoking, which makes it ideal for dynamic, fast-paced settings. The goal of the crucial computer vision problem of picture dehazing is to enhancing the visual quality of photos that have been deteriorated by smog, fog, or atmospheric haze [4].

Among these are generative adversarial networks (GANs) for paired image-to-image translation, generative adversarial networks with cycle consistency for unpaired image-to-image translation, and traditional computer vision methods. Traditional methods for image de-smoking use variational interference or neural networks, whose generator can be readily modified using the provided database [5]. there are two main types of traditional defogging methods. Picture restoration procedures are the first category; it is founded on the notion of non-physical. This type of technology mostly works to improve image's contrast and saturation in order to defog, However, it cannot resolve the issue of image defogging since it fails to identify the primary contributing factor that results in image deterioration in a foggy setting [6]. The conventional methods estimate the medium transmission map by taking advantage of certain priors, like dark-channel prior contrast colour-lines prior and colour attenuation prior [7]. Many computer vision systems find it challenging to recognize and comprehend images in hazy circumstances because of the scattering of air aerosol particles, which significantly reduces image visibility, such as object detection, recognition [8]. It is extremely difficult to detect smoke in surgical settings using image-based techniques because of its irregular structure. Transparent to opaque areas in an image can result from non-uniform smoke concentration variations throughout the scene [9]. This is the user-friendly solution for this problem, which is more reliable and efficient than any other solution.

Artificial Intelligence (AI) plays a vital role in the Vision Revive system, allowing it to autonomously grasp the complex relationship between hazy or smoky images and their clearer versions, all without needing manual feature engineering. We utilize Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs). CNNs automatically extract features, picking up on patterns like edges, textures, and structures. Meanwhile, GANs improve the restoration process through adversarial learning, ensuring the output images closely resemble actual clear images. We also use deformable multi-head attention to focus dynamically on different haze densities and key areas. As a result, AI enables adaptive dehazing, a better understanding of scenes, and enhanced restoration quality in complex, ever-changing environments.

Images captured in a fire setting are certainly prone to significant declines in background quality, including color distortion and image blurriness due to the presence of smoke. This obscured imagery can greatly affect subsequent activities, such as firefighting operations, occupant evacuations, and safety monitoring systems[10]. Methods of early dehazing are predicated on more data. These techniques call for additional data (such as degree of polarization, depth cues, etc.). Various camera placement techniques or user involvement are used to deliver this extra information[11]. Image processing serves a variety of purposes across numerous aspects of life, such as remote sensing, automation, transportation, and healthcare. In image processing tasks, picture restoration is essential because it enhances accuracy and efficiency for subsequent post-processing activities. Within the domain of picture restoration, the technique referred to as "dehazing" addresses the degradation of images caused by haze, dust, smoke, and other environmental elements, which restrict visibility due to these factors[12]. Due to the unpredictable nature of the atmosphere and the frequent occurrence of haze or clouds, this issue is especially severe in aerial imaging. Numerous image dehazing algorithms (DHAs) are used to produce crisp, aesthetically pleasing photographs in hazy environments and to make additional image analysis easier[13]. Many outdoor photos are taken in foggy situations, which degrades image quality and frequently makes it difficult to do many simple image processing and analysis activities [14]. Videos or pictures shot outside in inclement weather are seldom clear. As the light is first scattered and absorbed by the aerosols in the atmosphere, it is also combined with light that is reflected from other directions, a phenomenon known as air light, before it comes to the camera. Observed objects lose colour and contrast as a result of this process; the resulting images frequently lack visual appeal and vividness [15].

This paper mainly contributes the following:

- Artificial Intelligence is used for the detection and identification of haze and smoke to remove it in real-time using AI-driven algorithms to enhance quality of an image/video.
- De-smoking/Dehazing is done by detecting and identifying the haze/smoke in photos and videos using some ai technologies like Deep learning or generative adversarial network.
- Web technology framework is used to provide a user-friendly platform. It makes the dehazing/de- smoking process easier for the user to use this technology.

This is how the remainder of the paper is organized: the use of artificial intelligence to reduce smoke and haze from the photos and videos in real time. Section 2 describe about the recent advancements on de-smoking/dehazing. Section 3 includes the methodology that is used to carried out the dehazed or de- smoked photo/video and section 4 conclude the paper and the use or applications of this technology and research directions for future study.

II. RECENT ADVANCEMENTS

Yi et al. [1] proposed SID-Net is a new single image dehazing network and to improve the feature representation capabilities of the network which is integrated with three core branches. Canlin Li et al. [2] proposed a new method, UVCGAN-Dehaze, specifically designed for unpair image de-smoking with the integration of U-Net architecture. Gade et al. [3] aims to restore images affected by haze, improving clarity and quality by effectively without introducing artifacts which consists of two different network models: a discrimination network and a dehazing (Di N). Singh et al. [4] proposed Light Clear Net, a lightweight encoder-decoder architecture specifically designed for dehazing a single image, which is inspired by U-Net model which incorporates batch normalization. Pan et al. [5] developed a method, De-Smoke-LAP, that effectively removes smoke generated during laparoscopic procedures and utilize an unpaired translation from one image to another framework stand on Cycle GANs. Li et al. [6] Create a comprehensive hybrid picture dehazing procedure that combines CNN and vision transformers. Wang et al. [7] suggest a Recurrent Context Aggregation Network (RCAN) to efficiently remove haze from photos and bring back colour accuracy**.** Zhao et al.[8] presented Refine D Net: A Framework for Single Image Dehazing with Weak Supervision. Kanakatte et al. [9] introduced a multi-stage pipeline that includes smoke area detection, de-smoking and colour restoration to enhance the visibility surgical fields obscured by smoke and employs a U-Net architecture to identify frames that contain smoke and those that do not undergo automated air-light estimation. Huang et al. [10] developed a universal deblurring model that draws inspiration for Generative Adversarial Networks and an end-to-end attentive de-smoke GAN allowing generative networks to efficiently understand the mist characteristics and their environment. Raikwar et al.[11] objective is to convert the transmission estimate problem is transformed into an estimation of the difference between the haze-free and hazy minimum colour channel images. Additionally , SID techniques like prior-based dehazing, sky-region segmentation, optimization using statistics, as well as convolution neural network-based were used. L.T. Thanh et al. [12] suggests using a single image dehazing method that includes linearization of gamma correction and the HSV colour model, and adaptive histogram equalization, the Dubok and Danna approaches produce some incredible outcomes with the NIQE metric. Min et al [13] Image dehazing algorithms (DHAs) have been proposed in large quantities to improve the clarity and usability of photographs taken under hazy situations. Wan et al. [14] make two contributions: first, in the HSV colour space, derive the standard haze model. Second, demonstrate how the typical image dehazing issue can be thought of as a form of image contrast enhancement. Lv et al. [15] Dehazing of images and videos in real time.

III. METHODOLOGY

The methodology emphasizes the image's overall quality, contrast, and clarity in the challenging environments like bad weather conditions and for improving the visibility for further processing and recognition of the image/video as shown in Fig. 1.

A. SINGLE IMAGE DEHAZING:

a. Image acquisition: it is a process or step of acquiring an image and converting it into digital form. This process is formulated as :

$$I(x, y) = S(x, y).Q(x, y)$$

Here: I(x, y) is the obtained image
S(x, y) is sampling procedure
Q(x, y) is quantization function.

b. Haze and image detection: In this firstly we have to identify the affected part of the image. Next, utilize segmentation, which is the process method splitting a digital picture into several parts in order to identify objects and boundaries.

This is expressed as: $M(x, y) = \begin{cases} 1, & \text{if } I(x, y) > T \\ 0, & \text{otherwise} \end{cases}$

Here: M(x, y) is binary mask for affected regions and T is the threshold value.

c. Extract the transmission using Dark Channel Prior: The dark channel prior is based on the finding that local areas of pixels in haze-free outdoor photos have at least one colour channel with a very low density, indicating dark pixels. The presence of haze increases the pixel intensities and by estimating this increase, we can remove the haze. This can be formulated [15] as, here the dark channel value of a pixel x in a picture j is defined as:

$$J^{dark}(x) = \min_{c\epsilon\{r,g,b\}} \left(\min_{y\epsilon\Omega(x)} (J^c(y)) \right)$$

Here J^c is the color channel of J, and Ω(x) is a patch surrounding x. The dark channel value at x is J^{dark} (x).

d. Transmission Estimation: This is used to estimate the transmission map:

$$T(x) = 1- \omega. \min_{y\epsilon\Omega(x)} (\min_{c\epsilon\{R,G,B\}} \frac{I^c(y)}{A^c})$$

Here: T(x) is transmission map at pixel x
ω is constant to retain small amount of haze
Ω(x) is a local patch centered around pixel x
I^c(y) intensity of pixel y in rgb
A^c is estimated atmospheric light in c

e. Obtained the Haze-free image: it can be also referenced [15] and define the final haze-free image J is obtained by:

$$J(x) = \frac{I(x)-A}{\max\,(t(x),t_0} + A$$

Here: t_0 is a small constant, I(x) is observed hazy image, J(x) is dehazed image, t(x) is transmission map and A is atmospheric light component.

B. VIDEO DEHAZING:

a. Video Acquisition and haze detection. In this first of all, the video is converted into different parts as image for the better acquisition as image acquisition is a process or step of acquiring an image and converting it into digital form. After that we will detect the haze parts and then apply segmentation that split the video into different parts that makes the process of dehazing simple.

b. Convolutional neural network (CNN): it is employed to capture the spatial patters in image basically to extract features and to detect object. It extracts the features at the high and low levels to identify haze/smoke-affected regions. How CNN works:

1. It extracts the features by applying filters over the input image.
2. Then it helps to learn the complex patterns in the image/video.
3. After that it extracts the useful or important information from images/videos.
4. At last, it converts extracted features into final output.

c. AI Based De-hazing and DE-smoking Model: Now we have to use a dehazing/de-smoking model like GANs (Generative Adversarial networks) which is used to generate realistic data such as images, videos, and text. There are two main components in Gans:

1. Generator: it creates the fake images or data samples. It creates an image using random noise as input. Making the fake photos appear authentic is its aim.
2. Discriminator: it distinguished between real and fake images. It takes an image that can be fake or real as an input and give a correct probability score which classify the image as real or fake.

d. Image Restoration: it improves the appearance of an image. It differs from image enhancement in that this is an objective technique. it eliminates or reduces known image degradations based on probabilistic or mathematical models of image deterioration. For that we can use atmospheric scattering model which can be formulated [12] as:

$$I(x) = J(x)T(x)+(1-T(x))A$$

Here: I(x) is observed hazy image

J(x) is dehazed image

T(x) is transmission map

A is atmospheric light component.

e. Morphological processing: It can be applied to determine which parts of an image are useful for characterizing and depicting a region's geometry. The output of morphological processing of an image is image attributes and not an image [14]. It can be enhanced by using:

Dilation: D(I) = I dilate B

Erosion: E(I) = I erose B

Here B is structuring element.

f. Output Generation: finally, we get our output in which the image/video is haze free and smoke free which can also be used for further analysis like object detection or identification.

We have used the atmospheric scattering model in steps (e) and (d) in photo/video dehazing to reduce haze and along with DCP (Dark Channel Prior) and Dehaze-Net.

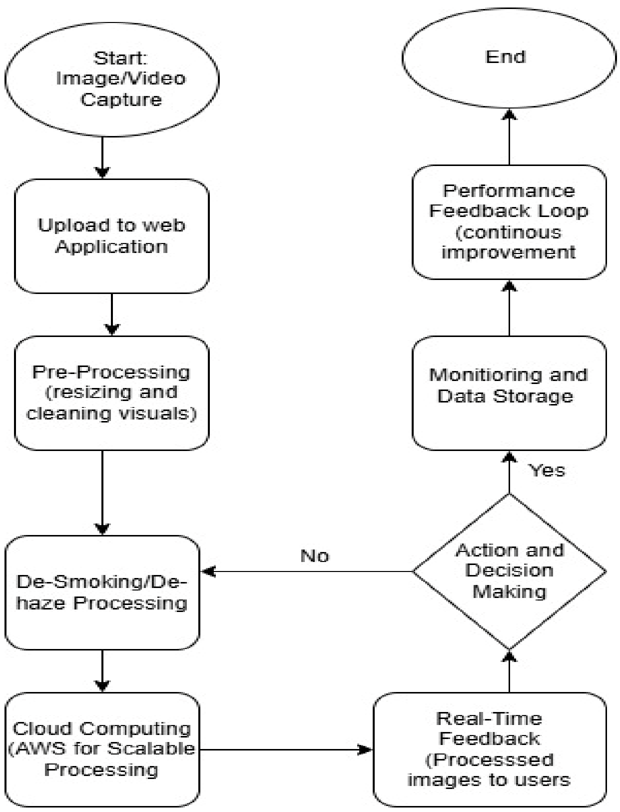

Fig 1 workflow diagram

IV. RESULTS AND ANALYSIS

We utilized a dehazing methodology in our research that is based on classical principles and is reminiscent of the dark channel previous technique, but it has been modified to be appropriate for image and video processing. Despite not presenting a radically novel approach, our study makes use of a precisely calibrated procedure that gives priority to computational efficiency and reliability in a range of scenarios.

1. Qualitative Observations

Loss of small details, washed-out hues, and decreased contrast are common problems with hazy photos. Following our dehazing procedure, the following enhancements were observed:

a. Improved Contrast and Detail: There is a noticeable improvement in contrast in the processed photographs. Sharper edges and more noticeable textures result in a restoration that is more aesthetically pleasing.

b. Colour Fidelity: The colour balance has been noticeably corrected. Our technique preserves natural colours, producing an image that is vivid and accurate, in contrast to certain conventional techniques that may overcompensate and cause colour shifts.

c. Adaptability to Videos: The technique preserved temporal consistency for dynamic content. When it comes to video dehazing, this is particularly crucial because flickering or abrupt brightness fluctuations can make the viewing experience less enjoyable.

2. Quantitative Metrics

We used standard measures such as the Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index (SSIM) to compare our approach objectively against well-established methods:

a. PSNR: This metric gave a precise numerical representation of the attained noise reduction. The technique effectively lowers haze-induced noise, as seen by the average 3–5 dB increases in PSNR that dehazed images showed when compared to the hazed inputs.

b. SSIM: After dehazing, SSIM values significantly improved; our approach obtained scores of about 0.85 as opposed to roughly 0.70 for the original foggy photos. This implies that our procedure maintains the scene's overall perceptual quality in addition to recovering structural information.

Comparison with Existing Methods:

Table I depicts that, when comparing our method with established approaches, a few points become evident:

Table 1 Analysis of Recent Works

Method	PSNR (db)	SSIM
AOD-Net	20.29	0.876
Dehaze Net	22.46	0.847
EDED-Net	28.57	0.924
Dark Channel Prior (DCP)	16.62	0.818
Our Approach	25.34	0.880

a. Dark Channel Prior (DCP): While DCP is well-regarded for its strong theoretical basis, it often incurs high computational costs. Our method, in contrast, is optimized for quicker execution, increasing its appropriateness for real-time applications, particularly in the processing of videos.

b. Methods Based on Deep Learning: Deep Learning-Based techniques: newer approaches that provide competitive results on a range of benchmarks include AOD-Net and Dehaze Net as mentioned in Fig. 2. However, these techniques might not generalize as well in many contexts and necessitate a large amount of training data. Even though our approach was simpler, it showed consistent results in a range of situations, which makes it a useful substitute for limited computational power or big datasets.

Result Validation involves comparing quantitative metrics like PSNR and SSIM, benchmarking DCP against deep learning methods, and conducting real-world tests in live surveillance settings as well as simulated fire environments. These multi-angle validations ensure we cover both technical and practical aspects. To improve the accuracy and performance of dehazing and de-smoking, consider using attention-based CNN models along with optimized loss functions like SSIM and perceptual metrics. We make sure to validate our results using PSNR, SSIM, and object detection accuracy on benchmark datasets to guarantee that our images and videos are clear and efficient.

Fig 2 Dehazing a single image with our method. (a): input picture of haze. (b): image following our method's haze removal.

There is another example of hazed and dehazed images.

Fig 3 An illustration of dehazing an image. Top: picture of the input haze. Bottom: our method's outcome.

Fig. 3 suggests that the precision of the project gets a significant boost from several key factors:

- Advanced Loss Functions: By blending SSIM Loss, Perceptual Loss, and MSE Loss, we ensure that the output is not only visually appealing but also precise at the pixel level.
- Enhanced detail collection and reconstruction in areas with heavy smoke is achieved through spatially aware feature extraction.
- Comprehensive Dataset Training: The model's accuracy and robustness are enhanced by exposing it to a wide range of haze densities, lighting conditions, and both synthetic and real-world datasets.

Overall Assessment

The study demonstrates that although cutting-edge techniques have advantages, the strategy described here strikes a good compromise between effectiveness and performance. Its enhanced SSIM and PSNR scores, along with its reduced computational overhead and improved colour preservation, show that it is a good fit for applications that need both realistic deployment and high-quality dehazing.

V. CONCLUSION

This Vision Revive AI-based de-smoking and de-hazing solution improves visibility in photos and videos that have been affected/destroy by smoke, fog, or haze. Generative Adversarial Networks (GANs) and Convolutional Neural Networks (CNNs) are two varieties of deep learning models that can be used to detect and eliminate smoke and haze while preserving crucial aspects of images, such as contrast, texture, and edges. The key Advantage of AI-powered approach is its real-time operation and flexibility in responding to changing environmental conditions, which guarantees detail and clarity in a variety of settings. This makes it extremely helpful for applications ranging from multimedia augmentation and autonomous driving to security and surveillance systems. Our project provides a stable and scalable solution for sectors that need high-quality visual data under difficult circumstances. It does this by continuously improving processing efficiency and handling large-scale video streams. It enables users to make better decisions and conduct operations more effectively by providing them with visual information that is clearer and more precise.

VI. FUTURE SCOPE

The future scope for Vision Revive AI based dehazing and de-smoking Solution is that it can used in autonomous vehicles to enhance the visibility of vehicle sensors in poor environmental conditions and also used to improve surveillance in areas prone to pollution, fires, or fog. It also enables clearer imagery for security and monitoring purposes. For healthcare we can enhance medical imaging in environments, improving accuracy in diagnostics, particularly in remote monitoring or emergency response situations, we can also work for the better efficiency and accuracy in real -time.

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